

IMS Hands-On Lab

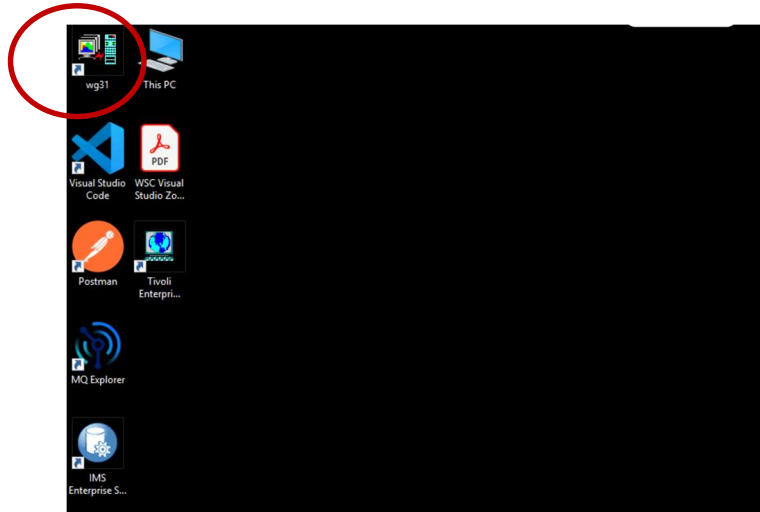
z/OS Connect and IMS OpenAPI 2

First things first

In this lab, the IMS Phonebook sample application (IVTNO) is used as the target IMS application program to expose through an API

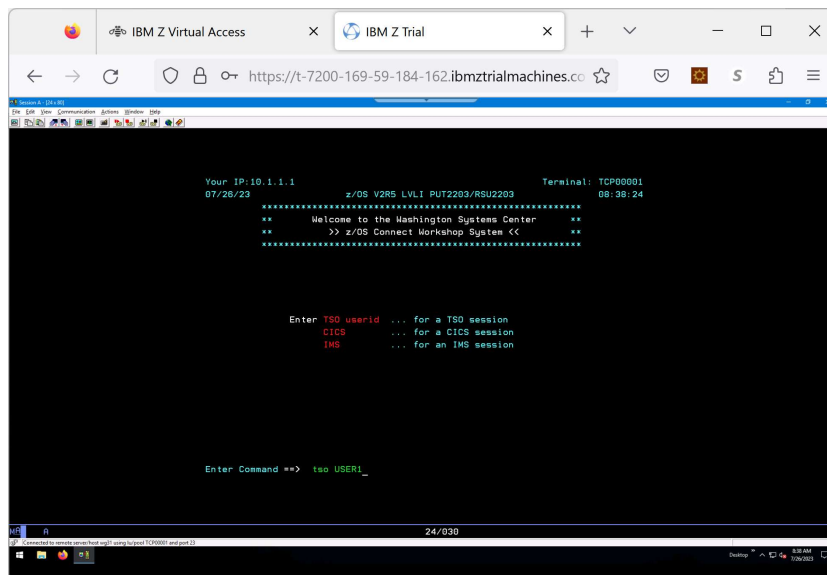
REVIEW the IMS Environment

Check out the IMS environment by double-clicking the **wg31 pcomm** icon

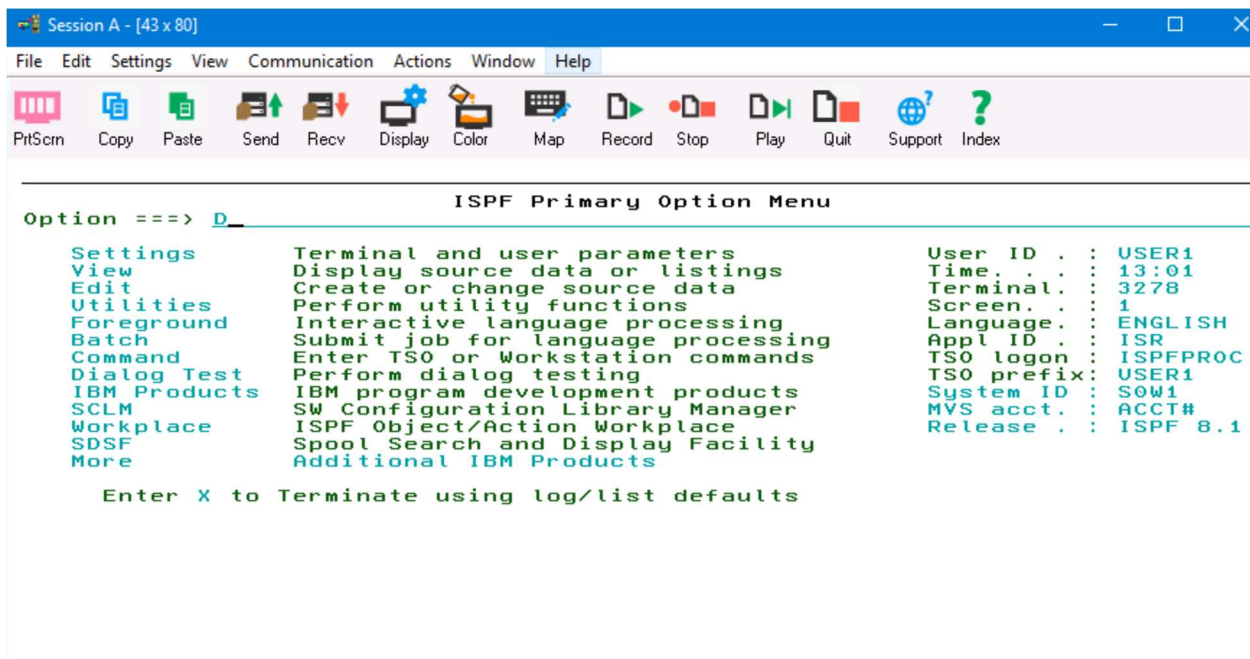


(note that shift- Enter is used as the enter function)

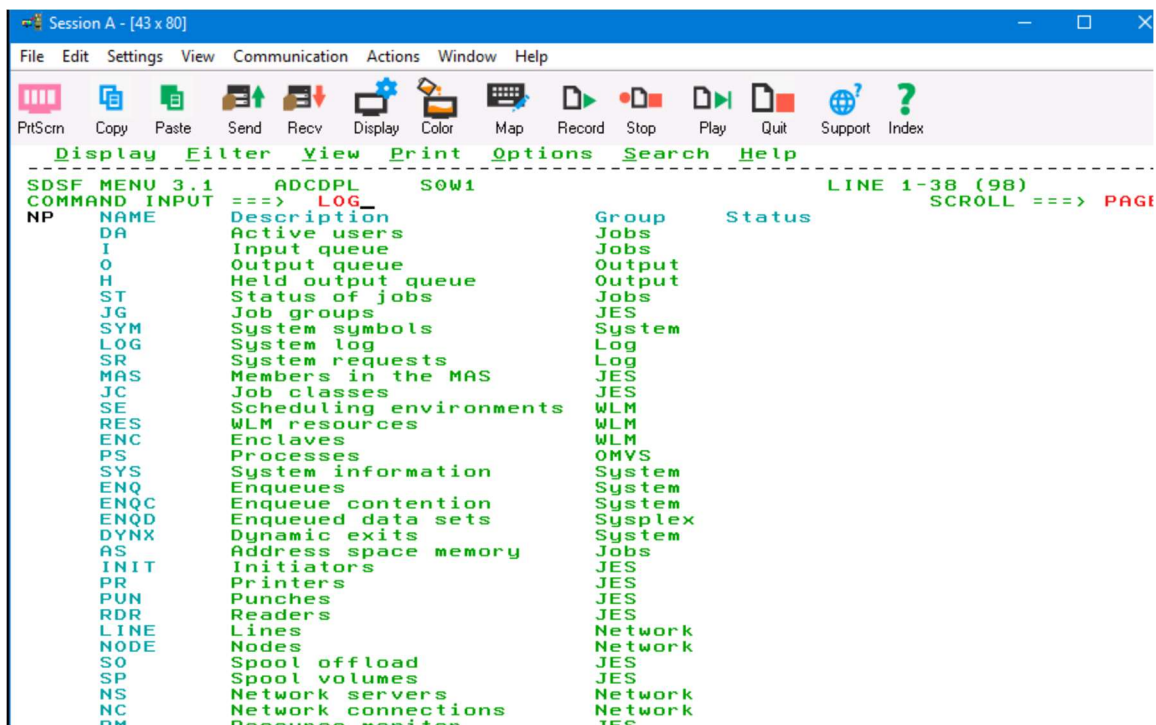
- Logon to TSO using USER1
 - Password is USER1



- ISPF option D allows access to SDSF



- Go to LOG



You will see the outstanding prompts for IMS (IVP1) and IMSConnect (IMS15HWS)

- Answer the outstanding prompt number (*nn*) for IMS (IVP1) to check on message region availability and the transaction ENQCT:

■ **/nn/DIS TRAN IVTNO.**

```

SDSF SYSLOG 0.101 SOW1 SOW1 04/30/2025 2W 6,744 COLUMNS 02- 81
COMMAND INPUT ==> /03/dis tran IVTNO_
*****
S FFFFFFF SOW1 25120 13:19:32.28 STC08771 00000090 IAZH1288 Operator verify
M 00000000 SOW1 25120 13:20:13.44 STC08771 00000090 +CSQX0041 QMZ1 CSQXSPRM
E 00000000 SOW1 25120 13:20:13.44 STC08771 00000090 used 34MB, free 1478MB:
M 00000000 SOW1 25120 13:20:13.54 STC08899 00000281 IRR8121 PROFILE * (G) IN
N 00000000 SOW1 25120 13:20:13.56 STC08899 00000281 TO START BPXAS W
N 00000000 SOW1 25120 13:20:13.56 STC08899 00000281 $HASP100 BPXAS ON STC
S 00000000 SOW1 25120 13:20:13.58 STC08899 00000281 $HASP373 BPXAS STARTE
M 00000000 SOW1 25120 13:20:13.58 STC08899 00000281 BPXP0241 BPXAS INITIATOR
E 00000000 SOW1 25120 13:20:13.58 STC08899 00000281 0062
M 00000000 SOW1 25120 13:20:13.58 STC08899 00000281 IRR8121 PROFILE * (G) IN
N 00000000 SOW1 25120 13:20:13.59 STC08900 00000281 TO START BPXAS W
N 00000000 SOW1 25120 13:20:13.60 STC08900 00000281 $HASP100 BPXAS ON STC
N 00000000 SOW1 25120 13:20:13.60 STC08900 00000281 $HASP373 BPXAS STARTE
S 00000000 SOW1 25120 13:20:13.61 STC08900 00000281 BPXP0241 BPXAS INITIATOR
M 00000000 SOW1 25120 13:20:13.68 STC08900 00000281 00B1
N 00000000 SOW1 25120 13:20:13.69 STC08900 00000281 IEF1961 IEF2371 0A80 ALL
N 00000000 SOW1 25120 13:20:13.69 STC08900 00000281 IEF1961 IEF2851 SYS1.L
M 00000000 SOW1 25120 13:20:13.69 STC08900 00000281 IEF1961 IEF2851 VOL SE
E 00000000 SOW1 25120 13:20:13.73 STC08900 00000281 IRR8121 PROFILE * (G) IN
M 00000000 SOW1 25120 13:20:13.74 STC08901 00000281 TO START BPXAS W
N 00000000 SOW1 25120 13:20:13.76 STC08901 00000281 $HASP100 BPXAS ON STC
N 00000000 SOW1 25120 13:20:13.81 STC08901 00000281 $HASP373 BPXAS STARTE
S 00000000 SOW1 25120 13:20:13.81 STC08901 00000281 BPXP0241 BPXAS INITIATOR
M 00000000 SOW1 25120 13:20:13.81 STC08901 00000281 00B4
N 00000000 SOW1 25120 13:20:29.49 TSU08898 00000281 IK1001 USERID USER1
N 00000000 SOW1 25120 13:20:29.49 TSU08898 00000281 IK1221 IPADDR..PORT 10.
N 00000000 SOW1 25120 13:20:29.49 TSU08898 00000281 IEA9891 SLIP TRAP ID=X62
N 00000000 SOW1 25120 13:20:29.49 TSU08898 00000281 IEA9891 SLIP TRAP ID=X13
N 00000000 SOW1 25120 13:20:29.49 TSU08898 00000281 IEA9891 SLIP TRAP ID=X13
N 00000000 SOW1 25120 13:20:29.51 TSU08898 00000281 IEF4501 USER1 ISPFPROC 1
N 00000000 SOW1 25120 13:20:29.53 TSU08898 00000281 $HASP395 USER1 ENDED
N 00000000 SOW1 25120 13:20:29.53 TSU08898 00000281 IEA9891 SLIP TRAP ID=X33
N 00000000 SOW1 25120 13:21:50.25 INSTREAM 00000281 LOGON
N 00000000 SOW1 25120 13:21:55.83 TSU08902 00000281 $HASP100 USER1 ON TSO
N 00000000 SOW1 25120 13:21:55.87 TSU08902 00000281 $HASP373 USER1 START
M 00000000 SOW1 10.19.33 STC08796 *03 DFS31391 IMS INITIAL I20, AUTOMATIC RESTA
S 00000000 SOW1 10.19.29 STC08796 *02 HWSC00001 *IMS CONNECT READY* IMS15HWS
*****
***** BOTTOM OF DATA *****

```

- Make note of the value for ENQCT which will keep track of the number of accesses against the IVTNO transaction.

```

SDSF SYSLOG 0.101 SOW1 SOW1 04/30/2025 2W 6,783 COLUMNS 52- 131
COMMAND INPUT ==>
*****
0090 DFS0001 TRAN CLS ENQCT QCT LCT PLCT CP NP LP SEGSZ SEGN0
0090 PARLM RC IVP1
0090 DFS0001 IVTNO 1 0 0 65535 65535 1 1 1 0 0
0090 NONE IVP1
0090 DFS0001 PSBNAME: DFSIVP1 IVP1
0090 DFS0001 *25120/132231* IVP1
0090 *04 DFS996I *IMS READY* IVP1
S READY* IVP1
IMS CONNECT READY* IMS15HWS
*****
***** BOTTOM OF DATA *****

```

- You can also issue a display active command to verify that the dependent regions are activating

■ **/nn/DIS A.**

```

SDSF SYSLOG 0.101 SOW1 SOW1 04/30/2025 2W 6,806 COLUMNS 52- 131
COMMAND INPUT ==>
*****
0090 CLASS IVP1
0090 DFS0001 1 IVP1 1 IMSMSG TP WAITING
0090 DFS0001 JMBPRGN JMB NONE IVP1
0090 DFS0001 JMBPRGN JMB NONE IVP1
0090 DFS0001 BATCHREG BMP NONE IVP1
0090 DFS0001 FPRGN FP NONE IVP1
0090 DFS0001 2 IMS150DM DBT IMS150DM AVAILABLE
0090 IVP1
0090 DFS0001 IMS15RC1 DBRC IVP1 IVP1
0090 DFS0001 IMS15DL1 DLS IVP1 IVP1
0090 DFS0001 VTAM ACB OPEN -LOGONS ENABLED IVP1
0090 DFS0001 IMSLU=N/A/N/A APPC STATUS=DISABLED TIMEOUT= 0
0090 MAXC= 5000 IVP1
0090 DFS0001 OTMA GROUP=OTMAGRP STATUS=ACTIVE IVP1
0090 DFS0001 APPC/OTMA SHARED QUEUE STATUS - LOCAL=INACTIVE
0090 GLOBAL=INACTIVE IVP1
0090 DFS0001 APPC/OTMA SHARED QUEUES LOGGING=N IVP1
0090 DFS0001 APPC/OTMA RRS MAX TCBS - 40 ATTACHED TCBS - 2 QUEUED
0090 DFS0001 0 IVP1
0090 DFS0001 APPLID=IMS15 GRSNAME= STATUS=DISABLED
0090 IVP1
0090 DFS0001 TCPIP_GENIMSID= STATUS=DISABLED IVP1
0090 DFS0001 LINE ACTIVE-IN - 1 ACTIV-OUT - 0 IVP1
0090 DFS0001 NODE ACTIVE-IN - 0 ACTIV-OUT - 0 IVP1
0090 DFS0001 *25120/132634* IVP1
0090 *05 DFS996I *IMS READY* IVP1
S READY* IVP1
IMS CONNECT READY* IMS15HWS
*****

```

- Answer the outstanding prompt number (zz) for IMS Connect (IMS15HWS)

- Respond with /zzVIEWHWS

```

Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
SDSF SYSLOG 0.101 SOW1 SOW1 04/30/2025 2W 6,806 COLUMNS 01- 80
COMMAND INPUT ==> /02viewhws_ SCROLL ==> PAGE
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 CLASS IVP1 1 IMSMS
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I 1 IVP1 JIMPRG
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I JIMPRG
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I BATCH
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I FPRGN
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I 2 IMS15
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I IVP1 IMS15
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I IMS15
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I VTAM ACB OP
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I IMSLU=N/A.N
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I MAXC= 5000 IVP1
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I OTMA GROUP=
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I APPC/OTMA S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I GLOBAL=INACTIVE
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I APPC/OTMA S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I APPC/OTMA R
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 RRSWKS- 0 IVP1
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I APPLID=IMS1
S
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 IVP1
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I TCPIP_GENIM
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I LINE ACTIVE
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I NODE ACTIVE
N 4200000 SOW1 25120 13:26:34.71 STC08759 00000090 DFS000I *25120/1326
W 4200000 SOW1 25120 13:26:34.72 STC08759 00000090 *05 DFS996I *IMS READY*
4200000 SOW1 12.26.34 STC08759 *05 DFS996I *IMS READY* IVP1
8000000 SOW1 10.19.29 STC08796 *02 HWSC0000I *IMS CONNECT READY* IMS15HWS
***** BOTTOM OF DATA *****

```

- Make sure that Datastore = IVP1 status is active

```

Session C - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
SDSF SYSLOG 0.101 SOW1 SOW1 04/30/2025 2W 6,845 COLUMNS 52- 131
COMMAND INPUT ==> ADAPTER=N MAXCVRT=100 NUMCVRT=0
)090 HWSC0001I MAXLSSSZ=32767
)090 HWSC0001I ODBM AUTO CONNECTION=Y
)090 HWSC0001I ODBM TIMEOUT=6000
)090 HWSC0001I ODBM IMSPLEX MEMBER=IMS15HWS TARGET MEMBER=PLEX1
)090 HWSC0001I DATASTORE=IVP1 STATUS=ACTIVE
)090 HWSC0001I GROUP=OTMAGRP MEMBER=HWSMEM
)090 HWSC0001I TARGET MEMBER=OTMAMEM STATE=AVAIL
)090 HWSC0001I DEFAULT REROUTE NAME=HWS$DEF IMS VERSION=15.1
)090 HWSC0001I RACF APPL NAME=IMSAPPL MULTIRTP= CASCADE=
)090 HWSC0001I OTMA ACEE AGING VALUE=999999 SENDALTP= TRANORCH=
)090 HWSC0001I OTMA ACK TIMEOUT VALUE=120
)090 HWSC0001I OTMA MAX INPUT MESSAGE=5000
)090 HWSC0001I SUPER MEMBER NAME= CM0 ACK TOQ=
)090 HWSC0001I IMSPLEX=PLEX1 STATUS=ACTIVE
)090 HWSC0001I MEMBER=IMS15HWS TARGET=PLEX1
)090 HWSC0001I ODBM=IVP1500D STATUS=REGISTERED ODBMRRS=N
)090 HWSC0001I ALIAS=IVP1 STATUS=ACTIVE
)090 HWSC0001I NO ACTIVE MSC
)090 HWSC0001I NO ACTIVE ISC
)090 HWSC0001I PORT=4000 STATUS=ACTIVE KEEPAV=0 NUMSOC=1
)090 HWSC0001I EDIT= TIMEOUT=0
)090 HWSC0001I NO ACTIVE CLIENTS
)090 HWSC0001I PORT=5555D STATUS=ACTIVE KEEPAV=0 NUMSOC=1
)090 HWSC0001I EDIT= TIMEOUT=6000
)090 HWSC0001I NO ACTIVE CLIENTS
)090 HWSC0001I NO ACTIVE RMTIMSCON
)090 HWSC0001I NO ACTIVE RMTICIS
IMS CONNECT READY* IMS15HWS
3 READY* IVP1

```

- If PORT=4000 shows an inactive status, issue the command /zzSTARTPT 4000

It will be beneficial, prior to creating the API, to see what the transaction does and to test its use from a terminal (just to make sure everything is working).

Logon to IMS by opening up another pcomm session and double-clicking on the **wg31 Pcomm** icon on the the desktop

```
Your IP:10.1.1.1          Terminal: TCP00007
04/30/25                z/OS V3R1 LVL1 PUT2309/RSU2309    13:40:49

*****
**      Welcome to the Washington Systems Center      **
**      >> Workshop System <<                        **
*****

Enter TSO userid  ... for a TSO session
CICS             ... for a CICS session
IMS              ... for an IMS session

Enter Command ==>  IMS_
```

Userid and password are USER1/USER1

Once you are logged on, use format IVTNO (/FOR IVTNO)

```
Session E - [24 x 80]
File Edit Settings View Communication Actions Window Help
PrtScn Copy Paste Send Recv Display Color Map Record Stop Play Quit Support
/ FOR IVTNO _

DFS3650I SESSION STATUS FOR IMS IVP1

DATE: 04/30/25      TIME: 13:44:02
NODE NAME:          TCP00007    IPADDR: 10.1.1.1..50846
USER:               USER1
PRESET DESTINATION:

CURRENT SESSION STATUS:

OUTPUT SECURITY AVAILABLE
```

Test the transaction using a process code of **DISPLAY** and a LAST NAME of **LAST1** (by the way, this is what we will be exposing using an API)

```

Session E - [24 x 80]
File Edit Settings View Communication Actions Window Help
PrtScrn Copy Paste Send Recv Display Color Map Record Stop Play Quit Support Index

*****
*      IMS INSTALLATION VERIFICATION PROCEDURE      *
*****

TRANSACTION TYPE : NON-CONV (OSAM DB)
DATE              : 04/30/2025

PROCESS CODE  (*1) :   DISPLAY

LAST NAME      :   LAST1_

FIRST NAME     :

EXTENSION NUMBER :

INTERNAL ZIP CODE :

(*1) PROCESS CODE
ADD
DELETE
UPDATE
DISPLAY
TADD

SEGMENT# :
```

If everything on the IMS system is working properly, you will receive the results as follows

```

Session E - [24 x 80]
File Edit Settings View Communication Actions Window Help
PrtScrn Copy Paste Send Recv Display Color Map Record Stop Play Quit Support Index

*****
*      IMS INSTALLATION VERIFICATION PROCEDURE      *
*****

TRANSACTION TYPE : NON-CONV (OSAM DB)
DATE              : 04/30/2025

PROCESS CODE  (*1) :   _DISPLAY

LAST NAME      :   LAST1

FIRST NAME     :   FIRST1

EXTENSION NUMBER :   8-111-1111

INTERNAL ZIP CODE :   D01/R01

(*1) PROCESS CODE
ADD
DELETE
UPDATE
DISPLAY
TADD

ENTRY WAS DISPLAYED

SEGMENT# :   0001
```

HOW to encode your Userid/password for the API authentication

You won't need to do this for your lab since this has been done for you. We will use userid USER1 and password USER1.

This information is provided for your use in your own environment

Open up a browser and locate <https://www.base64encode.org/> and key in Userid:Password in the ENCODE window. Click ENCODE, and view the result in the bottom window. For our example,

The screenshot shows the website www.base64encode.org. The main heading is "Encode to Base64 format" with the instruction "Simply enter your data then push the encode button." Below this is a large text input area containing "USER1:USER1", indicated by an orange arrow. Underneath the input area are several options: "UTF-8" for the destination character set, "LF (Unix)" for the destination newline separator, and checkboxes for "Encode each line separately", "Split lines into 76 character wide chunks", and "Perform URL-safe encoding". There is also a "Live mode OFF" toggle. An orange arrow points to the green "ENCODE" button, which is labeled "Encodes your data into the area below." Below the button is the output area, which contains the encoded string "VVFNFUJE6VVFNFUJE=", also indicated by an orange arrow.

When asked for authorization in the SWAGGER UI, code: BASIC VVFNFUJE6VVFNFUJE=